

Pratik Bairoliya

pratik0520@gmail.com | linkedin.com/in/pbairol | github.com/pbairoliya | pbairoliya.github.io

EXPERIENCE

Capital One

Jul. 2026 – Present

Software Engineer II — End-of-Day, custom batch ledger

McLean, VA

- Own the nightly **End-of-Day settlement** for every Card, Bank, and Financial Service account.
- Leading **SSUI** integration into the EOD APIs, unblocking downstream data transformation in production.
- Building observability and metrics for a new **Kubernetes + Temporal** platform shipped on **Argo/Helm**.
- De-risking V1/V2 endpoint cutovers behind **Optimizely** feature flags, keeping rollbacks one toggle away.

Capital One

Aug. 2025 – Jul. 2026

Software Engineer I — End-of-Day, custom batch ledger

McLean, VA

- Took a **batch data persistence API** from zero to production in **Go** — now the ledger's system of record.
- Automated publication of **80M+ records** across **2M+ accounts** nightly via **Hive** tuning on **AWS EMR**.
- Closed a priority auth gap for Card and Bank accounts with a **PoP token** authentication integration.
- **Top performer** on an 8-person team, shipping **2×** the team average in H1 2026.

Capital One

May 2024 – Aug. 2024

Software Engineering Intern

Richmond, VA

- Rebuilt the military rewards catalog off SharePoint as a **Vue.js/Spring Boot/DynamoDB** app, solo.
- Cut service latency **185%** (\$9M projected revenue); owned the **Terraform/Docker/Jenkins** deploy.

Fidelity Investments

Jun. 2023 – May 2024

Software Engineer (Intern, then part-time)

RTP, NC

- Moved **45%** of mainframe stock and job requests to **AWS** with **Terraform**, serving **400+** associates.
- Modeled **10,000** mainframe API nodes in **Neo4j**, cutting **5,000+** duplicates via Cypher automation.
- Shipped the full stack — **Angular** front end, **Java** services, **S3** metadata sync — on a 4-person Agile team.

PROJECTS

Daily Lookback | *Python, Ollama (local LLM), Google APIs, SQLite, launchd*

2026

- Local-first daily briefing: Calendar, Gmail, and iMessage into one generated note, **100% on-device**.
- Wrapped nondeterministic LLM output in a deterministic rules layer so reruns are **idempotent**, not re-rolled.
- Append-only state stores make every scheduled run safely repeatable; end-to-end tested with all I/O stubbed.

Receipt Pipeline | *Python, Apple Vision OCR, Ollama, file watchers*

2026

- Scan-to-structured-data pipeline: OCR any receipt, classify it with a local LLM, emit expense reports.
- Fingerprint dedup over an append-only index as source of truth; **52 tests**, file-locked against concurrent runs.

lc | *Python, Git, CLI design*

2026

- CLI that reconciles Obsidian notes with a Git repo, with spaced repetition and a nightly auto-push; **94 tests**.

TECHNICAL SKILLS

Languages: Go, Python, Java, TypeScript/JavaScript, SQL, C/C++, Bash

Cloud & Infrastructure: AWS (EMR, S3, DynamoDB, EC2, IAM), Kubernetes, Temporal, Docker, Terraform, Argo, Helm, Jenkins, CI/CD

Data & Backend: Distributed batch processing, Hive, Spring Boot, REST API design, GraphQL, Neo4j, PostgreSQL, Node.js

Frontend: Vue.js, React, Angular, TypeScript, HTML/CSS

AI/ML: Local LLM inference (Ollama), prompt design, OCR pipelines, PyTorch, deterministic LLM output design

Practices: Distributed systems design, observability, performance testing, zero-downtime migrations, Agile

Certification: AWS Certified Solutions Architect – Associate (Mar. 2026)

EDUCATION

North Carolina State University

Aug. 2021 – May 2025

B.S. Computer Science and B.S. Applied Mathematics — GPA: 4.0/4.0

Raleigh, NC

- **Park Scholar** — full-ride merit award to the top **1%** of applicants for leadership and service.